

Car-Reversing Audio-Visual Alarm

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Here is a simple car-parking alarm circuit based on an AVR microcontroller and an ultrasonic module. The circuit will alert you while you are reversing your car for parking, if there is any obstacle, through an audio-visual alarm.

Circuit and working

Fig. 1 shows the circuit for the car-reversing alarm. It is built around ATmega328 AVR microcontroller (IC1), 5V regulator (IC2), ultrasonic module HC-SR04, piezo buzzer (PZ1) and a few other components.

ATmega328. The heart of the sys-

tem is the 28-pin Atmel's ATmega328, which is an 8-bit AVR microcontroller with 32kB Flash and 1024 bytes of data RAM. It has two 8-bit timers/counters, one 16-bit timer/counter, six PWM channels, 23 programmable I/O lines, programmable serial USART master/slave SPI serial interface, a 6-channel 10-bit ADC and on-chip analogue comparator.

Pins 16 and 17 of IC1 are connected to Trigger and Echo pins of the ultrasonic module. Pin 18 of IC1 is connected to the piezo buzzer. The buzzer starts beeping when distance between the car's rear and an obstacle is 25cm or less.

Pins 4 through 6 and 11 through 15



of IC1 are connected to LED1 through LED8, respectively, to indicate distance between the car and the obstacle (refer Table I).

A 16MHz crystal oscillator is connected to pins 9 and 10 of IC1 to provide basic clock frequency. Power-on reset is provided by the combination of resistor R10 and capacitor C5. Switch S1 is used for manual reset of the microcontroller.

Ultrasonic transceiver HC-SR04.

HC-SR04 ultrasonic transceiver module (Fig. 2) uses sonar to determine the distance of an object, like bats or dolphins do. It offers excellent non-contact range detection of 2cm to 400cm with high accuracy and stable readings in an



Fig. 2: Ultrasonic ranging module HC-SR04

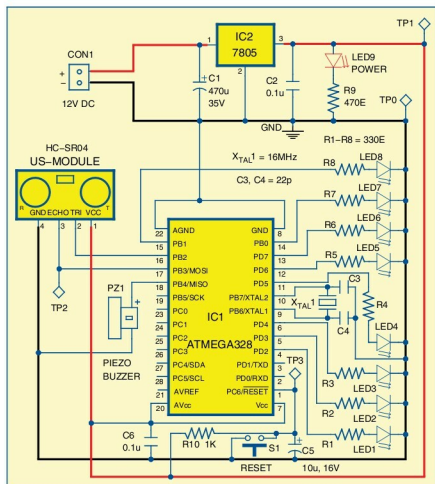


Fig. 1: Circuit diagram of the car-reversing audio-visual alarm

TABLE I
LED Indications

Approx. distance from obstacle	LEDs glowing
200-175cm	LED1
175-150cm	LED1-LED2
150-125cm	LED1-LED3
125-100cm	LED1-LED4
100-75cm	LED1-LED5
75-50cm	LED1-LED6
50-25cm	LED1-LED7
25-0cm	LED1-LED8